Ch. 3 Counting & Cardinality (Size)

$\bar{x}_2$
$\bar{y}_2$

☆ Def 1: If there is a bijection $\{1, \dots n\} \to S$ for some $n \in \mathbb{N}$, then we say S can be counted by n.

☆ Ex. $S = \{\Delta, O, \square\} = \{1, 2, 3\}$ I claim S can be counted by 3.

✓ Define the function $f: \{1,2,3\} \to S$ by $f(1) = \Delta$, $f(2) = \square$, $f(3) = O$
We can explicitly verify it is a bijection
Thus S is counted by n.

☆ Thm 1: (The invariance of counting) Suppose S can be counted by m & n; then $m = n$.

☆ Def 2: A set S is finite if there is some $n \in \mathbb{N}$ such that S is counted by n. The cardinality of S is the $n \in \mathbb{N}$ such that S is counted by n.
This is denoted $|S|$ or $\#$

☆ Prop 2. Let S be a unique function $\emptyset \to S$ ~~unique function~~ ~~unless~~ $S = \emptyset$
This is ~~inj~~ surjective
A function is ~~injective~~ surjective $\longleftrightarrow$ $S = \emptyset$ Conversely there are no functions $S \to \emptyset$ unless $S = \emptyset$

~~It is surjective $\longleftrightarrow$ $S = \emptyset$~~

A function from a set A to a set B is a rule that assigns to each element of A a unique element of B
$f: A \to B$ is injective if $\forall\, b \in B$ there exists at most one $a \in A$ s.t. $f(a) = B$
$f: A \to B$ is surjective if $\forall\, b \in B$ $\exists$ an $a \in A$ s.t. $f(a) = b$

hm 8. $|S| = 0 \iff S = \emptyset$ We suppose S is finite
Negating $\iff$: $|S| > 0$ if and only if $S \neq \emptyset$

Thm 9. Let $n \in \mathbb{N}$ the set $\{1, \dots n\}$ is finite and has cardinality n.

Proof: I need a bijection $\{1, \dots n\} \hookrightarrow \{1, \dots n\}$

Thm 10. If $|S| = n$ and $f: S \hookrightarrow T$ then $|T| = n$.

Proof $\{1, \dots n\} \hookrightarrow S \xhookrightarrow{f} T$

composing yields a bijection

Thm 11. If $|S| = |T| = n$, then there exists a bijection $S \hookrightarrow T$

Proof: $\{1, \dots, n\} \xhookrightarrow{h} S$, $\{1, \dots, n\} \xhookrightarrow{g} T$

bijective functions are invertible, so there is a bijection $h^{-1}: S \hookrightarrow \{1, \dots, n\}$